



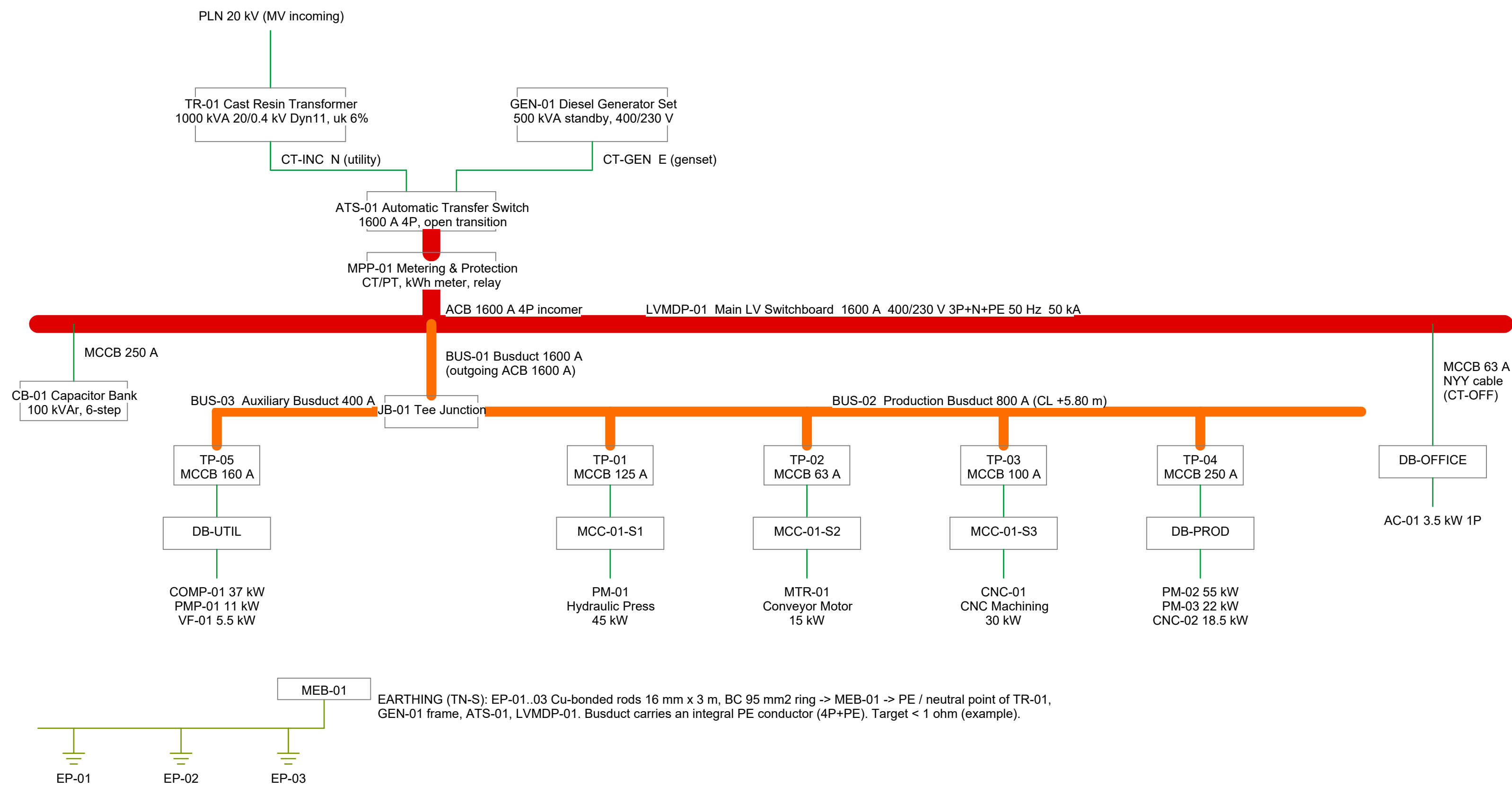
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LEGEND: Thick orange = busduct | Red = switchboard busbar | Green = cable / cable tray feeder | Olive = earthing conductor

NOTE: All ratings (kVA, A, kW, cable sizes) are TRAINING EXAMPLES. Final values require load calculation, voltage drop, short-circuit calculation and protection coordination to the applicable standards (e.g. PUIL 2020, IEC 60364, IEC 61439-6).

SLD-01 Single Line Diagram

(Simplified)

1 : 100

SYSTEM FLOW: PLN 20 kV -> TR-01 1000 kVA / GEN-01 500 kVA standby -> ATS-01 -> LVMDP-01 -> Main busduct BUS-01 (1600 A) -> JB-01 -> BUS-02 (800 A) / BUS-03 (400 A) -> Tap-off TP-01..05 -> MCC-01 / DB -> Industrial loads

COLOUR CODE (view filters): Orange = busduct | Yellow = tap-off | Red = TR, GEN, ATS, LVMDP, CB | Blue = MCC / DB | Green = feeder tray | Cyan = outgoing tray | Dark red = incoming tray | Olive = earthing | Purple = loads

SYSTEM: 400/230 V, 50 Hz, 3P+N+PE (TN-S). Revit distribution system "400/230V 3P4W 50Hz". Native power circuits: loads -> MCC/DB -> LVMDP-01 -> ATS-01 (panel schedules and load totals calculated by Revit). Busduct route recorded in circuit comments and Connected From/To parameters.

ALL RATINGS ARE TRAINING EXAMPLES, not the result of a final electrical design.

Latihan BIM / Portfolio

Industrial Facility - Busduct Distribution

Single Line Diagram & Legend

Project number	IND-BD-003
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Date 25-09-2026

Muhammad Sabiq Almuttadin

Checked by	Checker
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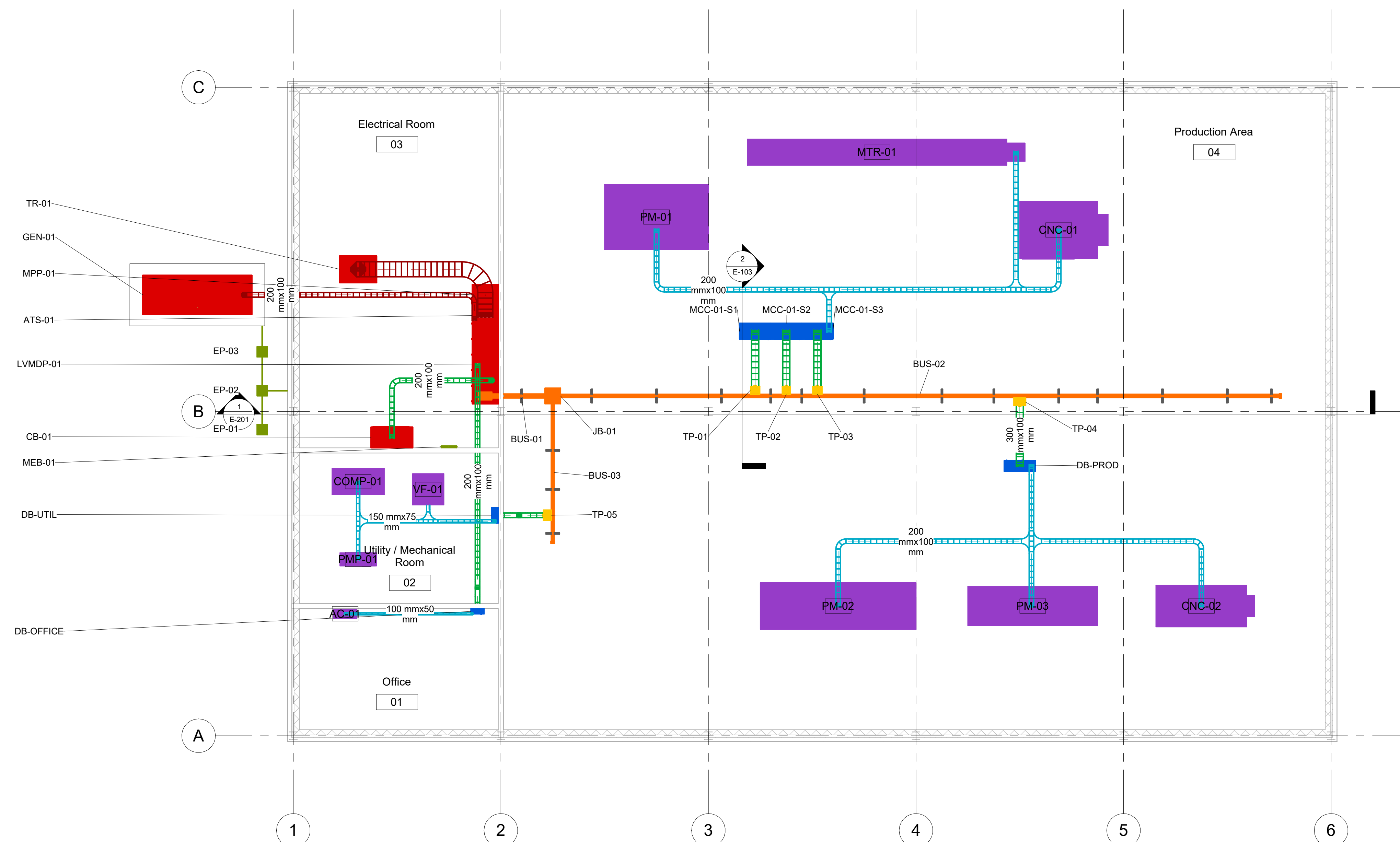
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Latihan BIM / Portfolio Industrial Facility - Busduct Distribution

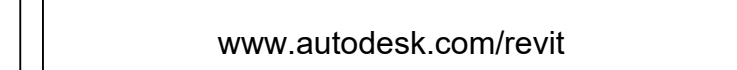
Electrical Layout Plan

Project number	IND-BD-003
Date	25-09-2026
Muhammad Sabiq Almuttaqin	
Checked by	Checker
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Scale	1 : 100

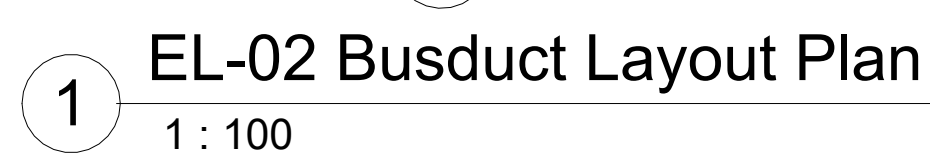
2019/00/2026	13	13	26
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1 EL-01 Electrical Layout Plan
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Industrial Facility - Busduct Distribution

Project number	IND-BD-003
Date	25-09-2026

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Scale	1 : 100
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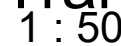
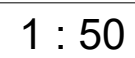
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Industrial Facility - Busduct Distribution

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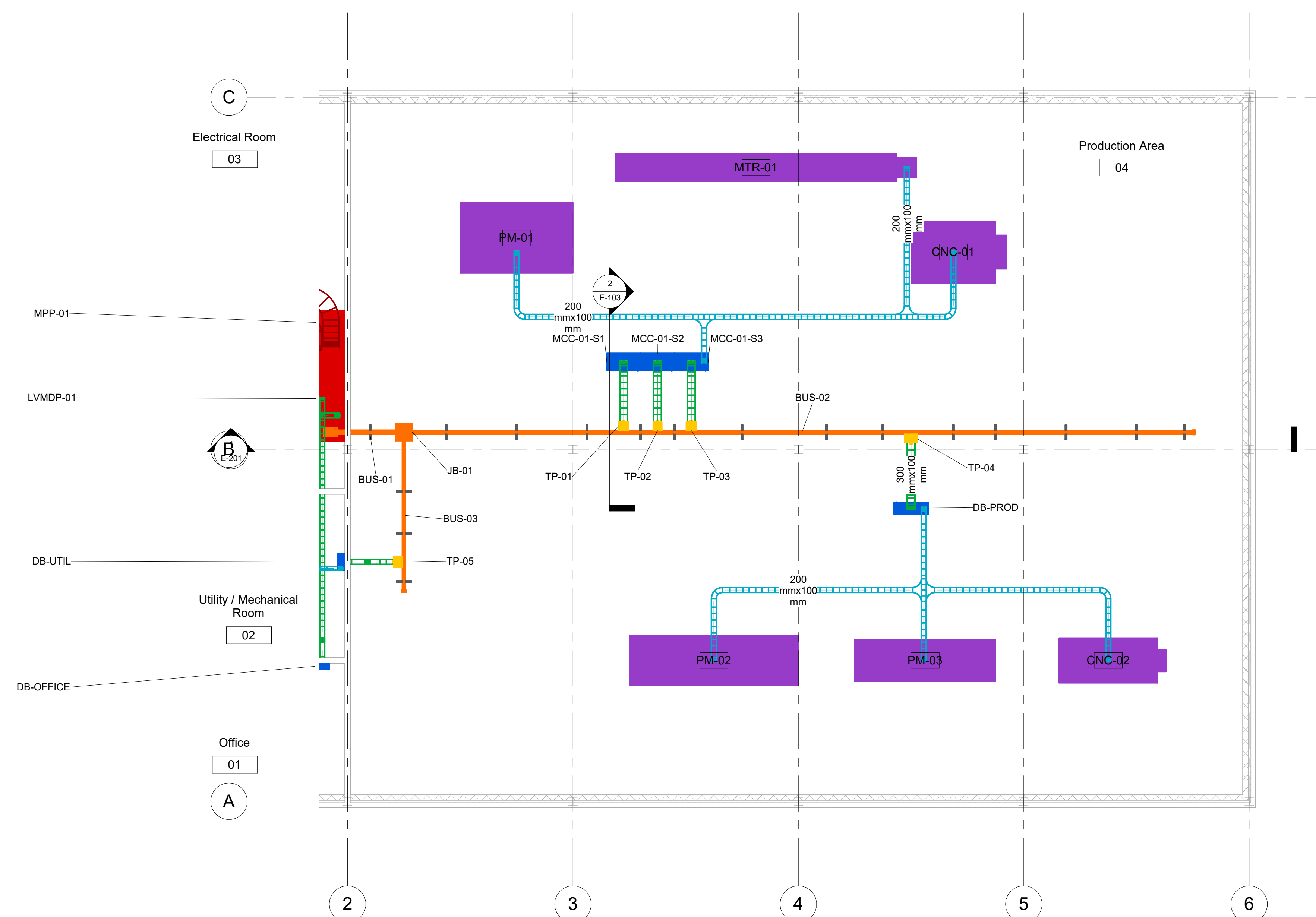
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1 EL-04 Production Area Plan
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Production Area Plan

Project number	IND-BD-003
Date	25-09-2026
Muhammad Sabiq Almuttaqin	
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Scale	1 : 100



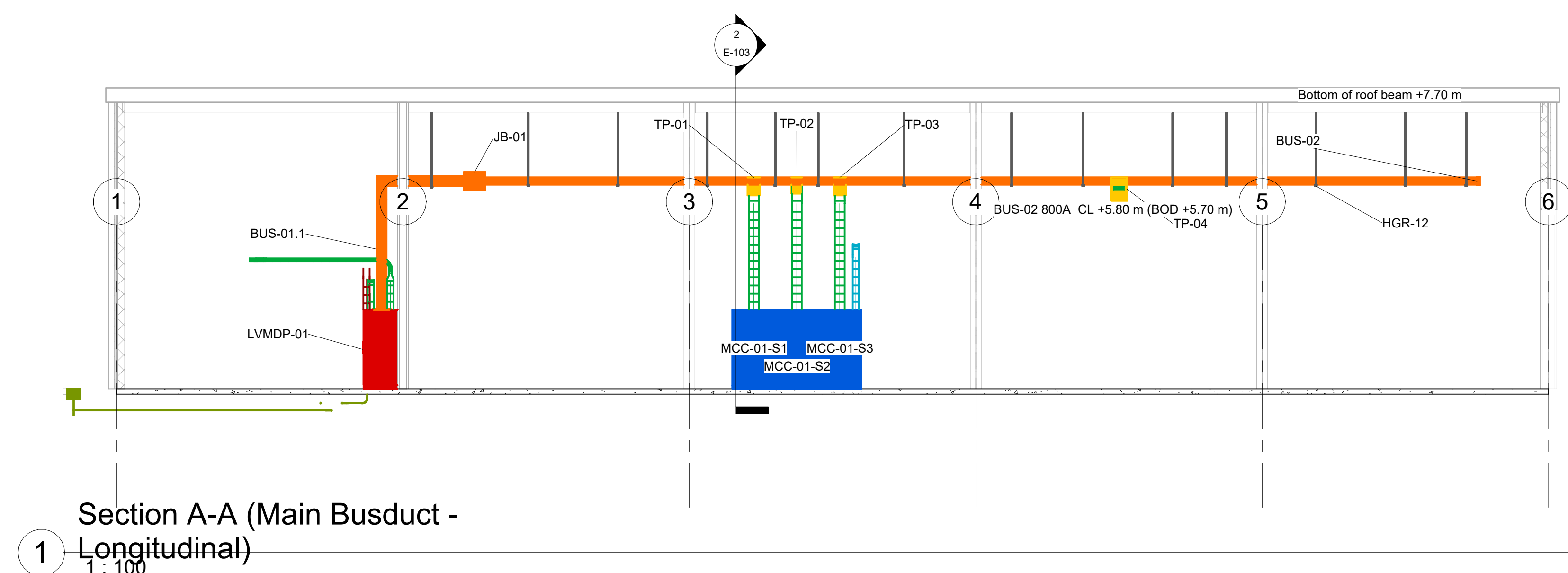
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Latihan BIM / Portfolio Industrial Facility - Busduct Distribution

Section A-A (Main Busduct)

Project number	IND-BD-003
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Date	25-09-2026
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Muhammad Sabiq Almuttaqin

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Tap-off Schedule					
Tap-off ID	Busduct	Rating	Connected Panel	Equipment	Area
TP-01	BUS-02	125A MCCB (example)	MCC-01-S1	PM-01 Hydraulic Press	Production Area
TP-02	BUS-02	63A MCCB (example)	MCC-01-S2	MTR-01 Belt Conveyor	Production Area
TP-03	BUS-02	100A MCCB (example)	MCC-01-S3	CNC-01 Machining Center	Production Area
TP-04	BUS-02	250A MCCB (example)	DB-PROD	PM-02, PM-03, CNC-02	Production Area
TP-05	BUS-03	160A MCCB (example)	DB-UTIL	COMP-01, PMP-01, VF-01	Production Area



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Industrial Facility - Busduct Distribution

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Panel Schedules DB

Date 25-09-2026

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Scale

Branch Panel: MCC-01-S1																					
Location: Production Area 04 Supply From: LVM DP-01 Mounting: Enclosure:								Volts: 400/230V 3P4W 50Hz Phases: 3 Wires: 4				A.I.C. Rating: Mains Type: Mains Rating: MCB Rating: 1 A									
Notes:																					
CKT	Circuit Description			Trip	Poles	A	B	C	Poles	Trip	Circuit Description			CKT							
1	PM-01			125 A	3	1763...								2							
3	--			--	--		1763...							4							
5	--			--	--			1763...						6							
7														8							
9														10							
11														12							
13														14							
15														16							
17														18							
19														20							
21														22							
23														24							
25														26							
27														28							
29														30							
31														32							
33														34							
35														36							
37														38							
39														40							
41														42							
				Total Load:		17633 VA	17633 VA	17633 VA													
				Total Amps:		77 A	77 A	77 A													
Legend:																					
Load Classification				Connected Load		Demand Factor		Estimated Demand		Panel Totals											
Other				52900 VA		100.00%		52900 VA													
										Total Conn. Load: 52900 VA											
										Total Est. Demand: 52900 VA											

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Branch Panel: MCC-01-S2																	
Location: Production Area 04					Volts: 400/230V 3P4W 50Hz					A.I.C. Rating:							
Supply From: LVMDP-01					Phases: 3					Mains Type:							
Mounting:					Wires: 4					Mains Rating:							
Enclosure:					MCB Rating: 1 A												
Notes:																	
CKT	Circuit Description			Trip	Poles	A	B	C	Poles	Trip	Circuit Description		CKT				
1	MTR-01			63 A	3	5867...							2				
3	--			--	--		5867...						4				
5	--			--	--			5867...					6				
7													8				
9													10				
11													12				
13													14				
15													16				
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31													32				
33													34				
35													36				
37													38				
39													40				
41													42				
				Total Load:		5867 VA	5867 VA	5867 VA									
				Total Amps:		26 A	26 A	26 A									
Legend:																	
Load Classification				Connected Load		Demand Factor		Estimated Demand		Panel Totals							
Other				17600 VA		100.00%		17600 VA									
										Total Conn. Load: 17600 VA							
										Total Est. Demand: 17600 VA							

Switchboard: LVMDP-01

Location:

Electrical Room 03

Supply From:

ATS-01

Mounting:

Enclosure:

Volts:

400/230V 3P4W 50Hz

Phases:

3

Wires:

4

A.I.C. Rating:

Mains Type:

Mains Rating:

MCB Rating:

1 A

Notes:

CKT	Circuit Description	# of Poles	Frame Size	Trip Rating	Load	Remarks
1	MCC-01-S1	3	400 A	125 A	52900 VA	
2	MCC-01-S2	3	400 A	63 A	17600 VA	
3	MCC-01-S3	3	400 A	100 A	35300 VA	
4	DB-PROD	3	400 A	250 A	112400 VA	
5	DB-UTIL	3	400 A	160 A	62900 VA	
6	DB-OFFICE	3	400 A	63 A	4100 VA	
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						
Total Conn. Load:					285200 VA	
Total Amps:					412 A	

Legend:

Load Classification	Connected Load	Demand Factor	Estimated Demand	Panel Totals
Other	285200 VA	100.00%	285200 VA	Total Conn. Load: 285200 VA Total Est. Demand: 285200 VA Total Conn.: 412 A Total Est. Demand: 412 A

Notes:



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Panel Schedules MCC & LVMDP

Date	25-09-2026
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Scale